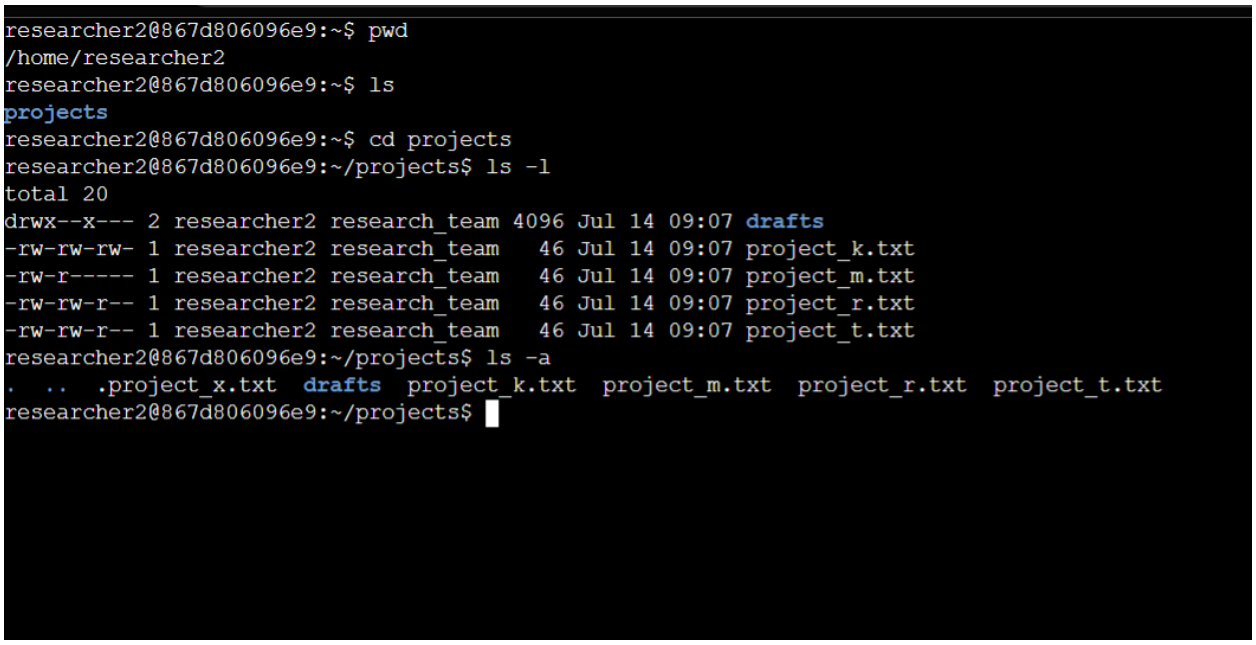
File permissions in Linux

Project description

[I accomplished how to list directory/file permission for files and hidden files and also change permissions depending on access.]

Check file and directory details



Describe the permissions string

[taking the drafts subdirectory permission, the 10 character string here indicates that d is a directory, characters 2-4 indicates that the user has read,write,execute permissions. Characters 5-7 shows that the group has only execute permission. And finally characters 8-10 show that the other has no permissions.]

Change file permissions

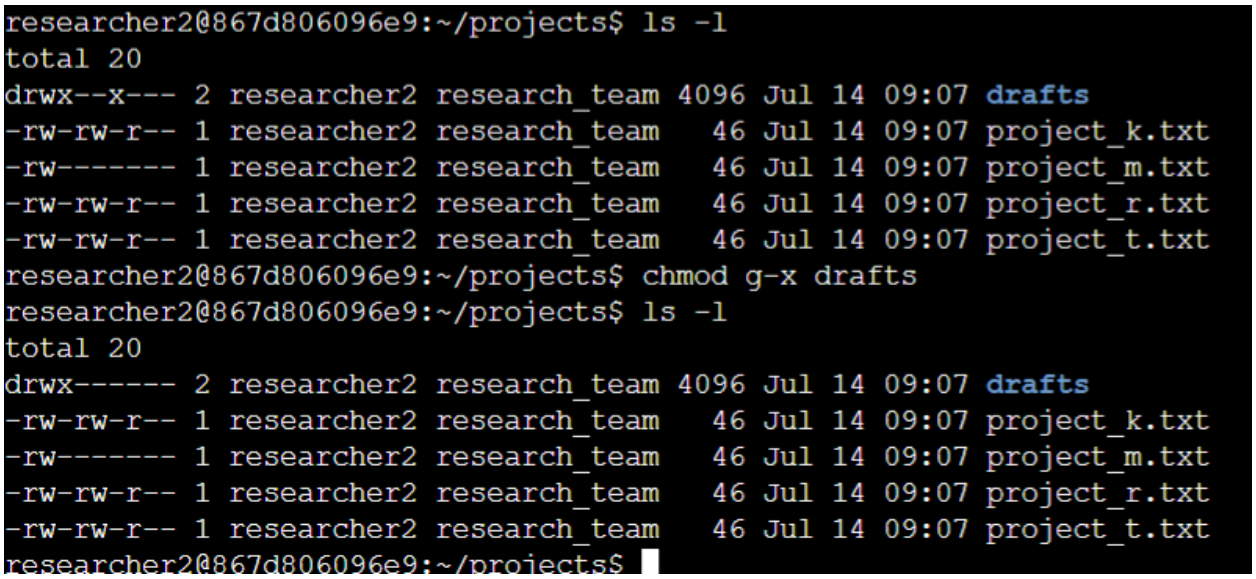
```
researcher2@867d806096e9:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jul 14 09:07 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_k.txt
-rw------- 1 researcher2 research_team   46 Jul 14 09:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_t.txt
researcher2@867d806096e9:~/projects$ chmod g-x drafts
researcher2@867d806096e9:~/projects$ ls -l
total 20
drwx------ 2 researcher2 research_team 4096 Jul 14 09:07 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_k.txt
-rw------- 1 researcher2 research_team   46 Jul 14 09:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_t.txt
researcher2@867d806096e9:~/projects$
```

Change file permissions on a hidden file

```
.  ..  .project_x.txt  drafts  project_k.txt  project_m.txt  project_r.txt  project_t.txt
researcher2@867d806096e9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 14 09:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 14 09:20 ..
-rw--w---- 1 researcher2 research_team   46 Jul 14 09:07 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jul 14 09:07 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_k.txt
-rw------- 1 researcher2 research_team   46 Jul 14 09:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_t.txt
researcher2@867d806096e9:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@867d806096e9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 14 09:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 14 09:20 ..
-r--r----- 1 researcher2 research_team   46 Jul 14 09:07 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jul 14 09:07 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_k.txt
-rw------- 1 researcher2 research_team   46 Jul 14 09:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 14 09:07 project_t.txt
researcher2@867d806096e9:~/projects$
```

Command used to change permission on a hidden file: chmod u-w,g-w, g+r .project_x.txt

Change directory permissions



Summary

[I accomplished how to list permission for directories and files and hidden files using the ls -l, ls -a, ls -la commands. Also remove or add permission to a certain user/group/other using the chmod command.
In the previous tasks some files/hidden files and a directory required a change in permission so that only authorized users with permissions can read, write or execute files/directory.]